E68: Localized Gradient-Energy Estimate

Counterexamples, Dimensional Classification, and Corrected Theorems

Mathematical audit of WCT Manuscript 4

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Abstract

The proposed E68 estimate bounds the gradient energy inside a ball by $CR^{n-2}\,\|\psi\|^2_{H^1(\mathbb{R}^n)}$. We prove that, with a constant independent of radius and field, this assertion holds for all H^1 fields and all radii exactly when $n = 2$. For every $n > 2$, a single compactly supported, singular H^1 field violates the estimate at arbitrarily small radii; a family of smooth compactly supported fields also rules out a universal constant. In dimension $n = 1$ the proposed estimate fails at large radii. We then prove a corrected inequality under the assumption $\nabla\psi \in L^n$ and obtain its H^2 consequence in dimensions three and four. These results concern analytic energy estimates, not the existence or dynamical stability of confined wave solutions.

1 Statement and quantifiers

For $n \geq 1$, let $B_R(x_0) = \{x \in \mathbb{R}^n : |x - x_0| < R\}$, and use the inhomogeneous norm

$$\|\psi\|^2_{H^1(\mathbb{R}^n)} = \|\psi\|^2_{L^2(\mathbb{R}^n)} + \|\nabla\psi\|^2_{L^2(\mathbb{R}^n)}\,.$$

The E68 assertion, interpreted as a uniform estimate, is

$$\boxed{\int_{B_R(0)} |\nabla\psi(x)|^2\,dx \leq C_n R^{n-2}\,\|\psi\|^2_{H^1(\mathbb{R}^n)}\,, \qquad \psi \in H^1(\mathbb{R}^n),\ R > 0.} \tag{E68}$$

Here $C_n < \infty$ must be independent of R and ψ. The same conclusions hold if one requires the estimate uniformly over all centers $x_0 \in \mathbb{R}^n$.

Theorem 1.1 (Exact dimensional classification)**.** *For the class $H^1(\mathbb{R}^n)$ and all radii $R > 0$, a uniform constant in* (E68) *exists if and only if $n = 2$. In that dimension $C_2 = 1$ works. For each $n > 2$, there is even a single $\psi \in H^1(\mathbb{R}^n)$ for which no finite constant depending on ψ can make* (E68) *hold for all sufficiently small R.*

Proof. We consider the three cases.

Case $n = 2$. Since $R^{n-2} = 1$ and $B_R \subset \mathbb{R}^2$,

$$\int_{B_R} |\nabla\psi|^2\,dx \leq \int_{\mathbb{R}^2} |\nabla\psi|^2\,dx \leq \|\psi\|^2_{H^1(\mathbb{R}^2)}\,.$$

Thus the inequality holds with $C_2 = 1$.

Case $n = 1$. Choose a nonconstant $\psi \in C_c^\infty((-1, 1))$ and put $A = \int_{\mathbb{R}} |\psi'|^2\,dx > 0$. For every $R \geq 1$,

$$\int_{-R}^{R} |\psi'|^2\,dx = A.$$

If (E68) were true, then

$$A \le C_1 R^{-1} \|\psi\|^2_{H^1(\mathbb{R})} \qquad (R \ge 1).$$

Sending $R \to \infty$ makes the right-hand side tend to zero, a contradiction.

Case $n > 2$**.** Fix

$$0 < a < \frac{n-2}{2},$$

and choose $\chi \in C_c^\infty([0,\infty))$ with $\chi(r) = 1$ for $0 \le r \le 1$ and $\chi(r) = 0$ for $r \ge 2$. Define, for $x \ne 0$,

$$\psi(x) = \chi(|x|)|x|^{-a},$$

with an arbitrary value at $x = 0$. Write $\sigma_{n-1} = |\mathbb{S}^{n-1}|$. Since $\psi = r^{-a}$ in B_1, polar coordinates yield

$$\int_{B_1} |\psi|^2 \, dx = \sigma_{n-1} \int_0^1 r^{n-1-2a} \, dr = \frac{\sigma_{n-1}}{n-2a} < \infty. \tag{1}$$

For $0 < |x| < 1$,

$$\nabla\psi(x) = -a|x|^{-a-2}x, \qquad |\nabla\psi(x)|^2 = a^2|x|^{-2a-2},$$

and therefore

$$\int_{B_1} |\nabla\psi|^2 \, dx = \sigma_{n-1} a^2 \int_0^1 r^{n-3-2a} \, dr = \frac{\sigma_{n-1}a^2}{n-2-2a} < \infty. \tag{2}$$

The cutoff-annulus contributions are finite, and ψ vanishes outside B_2. These classical derivatives on $\mathbb{R}^n\backslash\{0\}$ are also the weak derivatives: applying integration by parts outside B_ε leaves a boundary term bounded by a constant times ε^{n-1-a}, which tends to zero because $a < (n-2)/2 < n-1$. Thus $\psi \in H^1(\mathbb{R}^n)$.

For each $0 < R < 1$, the calculation in (2) gives

$$\int_{B_R} |\nabla\psi|^2 \, dx = \frac{\sigma_{n-1}a^2}{n-2-2a} R^{n-2-2a}. \tag{3}$$

Hence

$$R^{2-n} \int_{B_R} |\nabla\psi|^2 \, dx = \frac{\sigma_{n-1}a^2}{n-2-2a} R^{-2a} \longrightarrow +\infty \qquad (R \downarrow 0). \tag{4}$$

On the other hand, the proposed E68 bound would make the left-hand side at most $C_n \|\psi\|^2_{H^1} < \infty$, independently of R. This contradiction completes the proof. □

2 Failure on smooth fields

The preceding fixed-field counterexample has a singularity at the origin. The failure of a universal constant does not require any singular fields.

Proposition 2.1 (Smooth concentration counterexample)**.** *Let $n > 2$. For every $C > 0$, there are $\psi \in C_c^\infty(\mathbb{R}^n)$ and $R > 0$ for which*

$$\int_{B_R} |\nabla\psi|^2 \, dx > CR^{n-2} \|\psi\|^2_{H^1(\mathbb{R}^n)}.$$

Proof. Choose a nonconstant $\phi \in C_c^\infty(B_1)$ and put

$$\psi_R(x) = R^{1-n/2}\phi(x/R).$$

Let

$$A = \|\nabla\phi\|_2^2 > 0, \qquad B = \|\phi\|_2^2.$$

Differentiation and the substitution $y = x/R$ give

$$\|\nabla\psi_R\|_2^2 = A, \qquad \|\psi_R\|_2^2 = R^2 B.$$

Since $\operatorname{supp}\psi_R \subset B_R$,

$$\int_{B_R} |\nabla\psi_R|^2\,dx = A, \qquad \|\psi_R\|_{H^1}^2 = A + R^2 B.$$

A uniform E68 inequality would imply

$$A \le CR^{n-2}(A + R^2 B) \qquad \text{for every } R > 0.$$

The right-hand side tends to zero as $R \downarrow 0$, since $n > 2$, while $A > 0$ is fixed. This is impossible. □

3 Corrected estimate with critical gradient integrability

Theorem 3.1 (Corrected E68)**.** *Let $n > 2$ and suppose $\psi \in H^1(\mathbb{R}^n)$ with $\nabla\psi \in L^n(\mathbb{R}^n)$. For every $R > 0$ and $x_0 \in \mathbb{R}^n$,*

$$\boxed{\int_{B_R(x_0)} |\nabla\psi|^2\,dx \le \omega_n^{1-2/n} R^{n-2} \|\nabla\psi\|_{L^n(\mathbb{R}^n)}^2,} \tag{E68$'$}$$

where $\omega_n = |B_1(0)|$.

Proof. Apply Hölder's inequality on $B_R(x_0)$ with conjugate exponents $p = n/2$ and $q = n/(n-2)$:

$$\begin{aligned}
\int_{B_R(x_0)} |\nabla\psi|^2\,dx &= \int_{B_R(x_0)} |\nabla\psi|^2 \cdot 1\,dx \\
&\le \left(\int_{B_R(x_0)} |\nabla\psi|^{2p}\,dx\right)^{1/p} \left(\int_{B_R(x_0)} 1^q\,dx\right)^{1/q} \\
&= \left(\int_{B_R(x_0)} |\nabla\psi|^n\,dx\right)^{2/n} |B_R(x_0)|^{(n-2)/n} \\
&\le \|\nabla\psi\|_{L^n(\mathbb{R}^n)}^2 \,(\omega_n R^n)^{(n-2)/n} \\
&= \omega_n^{1-2/n} R^{n-2} \|\nabla\psi\|_{L^n(\mathbb{R}^n)}^2.
\end{aligned}$$

This is (E68′). □

Corollary 3.2 (Recovering the original H^1 right-hand side)**.** *Let $n > 2$ and let $\mathcal{A} \subset H^1(\mathbb{R}^n)$ be an admissible class satisfying the* independently established *a priori bound*

$$\|\nabla\psi\|_{L^n(\mathbb{R}^n)} \le K \|\psi\|_{H^1(\mathbb{R}^n)} \qquad \textit{for all } \psi \in \mathcal{A},$$

with K independent of ψ. Then

$$\int_{B_R(x_0)} |\nabla\psi|^2\,dx \le \omega_n^{1-2/n} K^2 R^{n-2} \|\psi\|_{H^1(\mathbb{R}^n)}^2 \qquad (\psi \in \mathcal{A}).$$

Proof. Insert the assumed gradient estimate into Theorem 3.1. □

Remark 3.3. *The hypothesis in Corollary 3.2 does not follow from H^1 membership alone. A governing PDE, a band limit, or an additional regularity restriction may supply such a bound, but it must be proved separately for the specified admissible class.*

4 An H^2 replacement in dimensions three and four

Corollary 4.1 (Higher-regularity localized estimate)**.** *If $n \in \{3, 4\}$ and $\psi \in H^2(\mathbb{R}^n)$, there is a constant $C_n < \infty$, independent of ψ, x_0, and R, such that*

$$\boxed{\int_{B_R(x_0)} |\nabla\psi|^2 \, dx \le C_n R^{n-2} \|\psi\|^2_{H^2(\mathbb{R}^n)} .} \tag{5}$$

Proof. When $n = 4$, the standard Sobolev inequality in $\mathbb{R}^4$ applied to $\nabla\psi \in H^1(\mathbb{R}^4)$ gives

$$\|\nabla\psi\|_{L^4} \le C \|\nabla\psi\|_{H^1} \le C \|\psi\|_{H^2} .$$

When $n = 3$, Sobolev embedding yields $\|\nabla\psi\|_{L^6} \le C \|\psi\|_{H^2}$. Interpolation between L^2 and L^6 gives

$$\|\nabla\psi\|_{L^3} \le \|\nabla\psi\|^{1/2}_{L^2} \|\nabla\psi\|^{1/2}_{L^6} \le C \|\psi\|_{H^2} .$$

In both dimensions the hypothesis of Theorem 3.1 holds with $\|\nabla\psi\|_{L^n} \le C \|\psi\|_{H^2}$. Its conclusion is precisely (5). □

Proposition 4.2 (The H^2 replacement fails above dimension four)**.** *If $n > 4$, there is no constant C_n for which (5) holds for all $\psi \in C_c^\infty(\mathbb{R}^n)$ and all $R > 0$.*

Proof. Take a nonconstant $\phi \in C_c^\infty(B_1)$ and set

$$u_R(x) = R^{2-n/2}\phi(x/R), \qquad 0 < R < 1.$$

Scaling gives

$$\|u_R\|_2^2 = R^4 \|\phi\|_2^2 , \quad \|\nabla u_R\|_2^2 = R^2 \|\nabla\phi\|_2^2 , \quad \left\|D^2 u_R\right\|_2^2 = \left\|D^2\phi\right\|_2^2 .$$

Thus $\|u_R\|^2_{H^2}$ stays bounded as $R \downarrow 0$, whereas

$$\int_{B_R} |\nabla u_R|^2 \, dx = R^2 \|\nabla\phi\|_2^2 .$$

If (5) held universally, it would imply

$$R^2 \|\nabla\phi\|_2^2 \le C R^{n-2}$$

for some fixed C and every small R. Dividing by R^2 gives $\|\nabla\phi\|_2^2 \le C R^{n-4} \to 0$, a contradiction. □

5 Status and scope for Manuscript 4

- **REFUTED:** E68 with its unrestricted H^1 hypothesis, when $n > 2$, even if the multiplicative constant is allowed to depend on one fixed singular H^1 field. The all-radii $n = 1$ version also fails.
- **PROVED:** E68 in dimension $n = 2$ with $C_2 = 1$.
- **PROVED:** Corrected E68′ when $n > 2$ and $\nabla\psi \in L^n$, with the displayed explicit constant.
- **PROVED:** The H^2-based estimate for $n = 3, 4$.
- **CONDITIONAL:** Recovery of the original H^1 right-hand side within a PDE-dependent admissible class, pending an independent uniform L^n gradient estimate.

These are purely analytic assertions. They do not establish existence, orbital stability, metastability, or a physical confinement-lifetime comparison in different spatial dimensions.

References

[1] *WCT Full Equation List (Corrected)*, E68 entry, `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/WCT_FULL_EQUATION_LIST_CORRECTED.md`.

[2] *WCT Lean Analytic Contracts*, definition `LocalEnergyBound`, `https://github.com/rickyjreyes/wct-lean/blob/main/WCTLean/Contracts/Analytic.lean`.